

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

Review of the Global Activity of Heart Transplant

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ABSTRACT: Heart failure is a global disease with significant morbidity. Heart transplant (HT) can be a lifesaving therapy for select patients with end-stage heart failure. In 2020, over 7000 HTs were performed globally; 90% of HTs were performed in the United States and Western Europe, with only 10% throughout the rest of the world. In this article, we offer an overview of the global landscape of HT, exploring challenges and prospects worldwide. We review HT practices, rates and post-HT outcomes, underscoring the differences between countries within each region. We review limitations hindering HT expansion, such as sociocultural factors, as seen in Japan and Israel; health care funding, in countries like India and South Africa; socioeconomic disparities in access, like the United States; and shortage in organ supply, as seen in China and Saudi Arabia. This review underscores the need to address limitations and highlights opportunities to enhance global HT accessibility, especially in lower- and middle-income countries.

Key Words: heart failure ■ heart transplantation ■ low and middle income countries ■ socioeconomic disparities

Heart failure (HF) is a complex syndrome with significant morbidity and mortality. In 2019, there were 56.2 million prevalent cases accounting for 9.91 million years lost to disability and 346 billion US\$ in expenditures worldwide, with the numbers projected to increase.^{1,2} Despite advances in HF therapeutics and wide implementation of guideline-directed medical therapy, a subset of patients with HF progress to advanced (stage D) HF, with a 75% mortality within 5 years of diagnosis.³ Heart transplant (HT) stands as a potential lifesaving therapy for those patients; however, its availability is limited, particularly in developing countries. There are numerous challenges to the wide implementation of HT. The availability of HT as a treatment is constrained by limited organ donor pools and the lack of robust transplant care worldwide.⁴ Advances such as donation after circulatory death (DCD) and ex vivo perfusion may expand the donor pool, less-invasive surveillance techniques (cell-free DNA) may allow for more accessible post-HT care, and international efforts

to strengthen regulatory bodies promise to increase HT accessibility.

In 2021, >7000 HTs were performed worldwide.⁵ The majority of HT, ≈90% between 2010 and 2018, took place in Europe and North America, while only 10% were performed in the rest of the world (Table 1; Figure 1).⁶ Precise figures, especially in developing nations, are not widely reported. Therefore, the World Health Organization along with Spain's National Transplant Organization collaborated to create the Global Observatory on Donation and Transplantation (GODT) to track organ transplants worldwide.⁵ The International Registry in Organ Donation and Transplantation is the first worldwide registry on organ donation/transplant and contains data from 93 countries worldwide since 1998.⁷ In this article, we aim to provide a summary review of the current state of HT globally, with an expanded focus on lower- and middle-income countries (LMIC). We will review adult HT activity, post-HT outcomes, country-specific regulations and challenges and explore future directions.

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Nonstandard Abbreviations and Acronyms

CRS	candidate risk score
CVD	cardiovascular disease
DCD	donation after circulatory death
GDP	gross domestic product
GODT	Global Observatory on Donation and Transplantation
HF	heart failure
HT	heart transplant
IABP	intra-aortic balloon pump
JOTN	Japan Organ Transplantation Network
LMIC	lower- and middle-income countries
MCS	mechanical support
PMP	per million population
SA	Saudi Arabia

METHODS

We conducted a narrative review of available adult HT data from select countries worldwide between January 2023 and March 2024.⁸ World regions were defined according to the Global Burden of Disease Study, and the 21 world regions were aggregated into the following 8 regions: High-Income North America, Latin America and the Caribbean, Western Europe, Central and Eastern Europe, the Middle East and North Africa, Sub-Saharan and South Africa, South Asia, East Asia and the Western Pacific.⁹ Countries were selected a priori based on expert opinion and public data availability. For each selected country when available, we extracted the number of HT performed, post-HT outcomes and listing criteria from national registries and public institutions that oversee HT activity. If data from national registries were not available, we utilized data from international registries, including the GODT, the International Registry in Organ Donation and Transplantation and the International Society for Heart and Lung Transplantation. In instances where neither national nor international registry data were accessible, we relied on single-center reports that reflect the country's experience. To obtain single-center data, we performed an English literature search using PubMed and Google Scholar for English-language articles. The World Bank, International Trade Administration and the Organization for Economic Co-operation and Development databases were used to obtain economic data.

HIGH-INCOME NORTH AMERICA

The United States

The United States has the highest health care expenditure in the world with nearly 17% of gross domestic product (GDP) spent on health care. Around 7 million Americans are living with HF, with ≈2.7% of patients having end-stage HF.¹⁰ The United States has one of the highest HT rates, with 12.4 HT recipients per million population (PMP) in 2022.⁵ The first adult HT in the United States was

performed in 1968.¹¹ In 1984, the Organ Procurement and Transplantation Network was established to regulate organ allocation. Organ Procurement and Transplantation Network contracts with the United Network for Organ Sharing, a nonprofit organization, to create an allocation system for HT that ensures a fair and equitable organ allocation.¹² In 2018 the Organ Procurement and Transplantation Network/United Network for Organ Sharing revised the HT allocation system to a 6-tiered allocation policy (Table 2).¹³

The Organ Procurement and Transplantation Network/United Network for Organ Sharing publishes an annual report with comprehensive data on HTs in the United States. The report provides a detailed update on HT activity, including waitlist outcomes, donation rate and post-HT outcomes data.¹⁴ Adult HT rates have steadily increased, from ≈2300 per year between 1995 and 2015 to >3500 in 2022.¹⁴ The HT rate in 2022 reached 122.5 transplants per 100 patient-years, the highest in a decade, an 82% increase since 2011.¹⁴ Over the same period, the number of donors increased; in 2022, 4223 deceased donor hearts were recovered for transplant. The largest increase in donors was especially notable among younger groups. For those aged 30 to 39 years, the number of donors increased by 194% over the past decade. The inclusion of donors positive for hepatitis C and the evolution of procurement techniques such as HTs using DCD partially explain the growth. Hepatitis C-positive donors increased by >11 thousand percent from 2011 and represented ≈10% of all donors in 2022. Similarly, DCD donors increased from 0.04% in 2011 to 8.8% in 2022. Unique to the United States, the opioid pandemic and gun-related deaths further expanded the donor pool. In 2017, >30% of donor-related deaths were related to opioid overdose or gun-related.¹⁵ The increase in donors was matched by an increase in the number of new candidates; 4447 candidates were added to the HT waitlist in 2022.¹⁴ The increase in donors has led to a decrease in waitlist time with nearly half of candidates waiting <90 days.¹⁴ However, despite the shorter wait times, pretransplant mortality remained stable at ≈9 deaths per 100 patient-years between 2019 and 2022. In 2022, 8% of HT candidates either died or were removed from the waitlist for being too sick for transplant.

The United States has favorable post-HT survival when benchmarked against European and non-European global data.⁶ Six-month post-HT mortality was 8.8% in 2014 and reached a nadir of ≈7% in 2018. One-year post-HT mortality remained stable at 9% to 10% over the past decade. By comparison, 1-year mortality after HT in Europe between 2012 and 2017 was 19%, and 18% in non-European centers reporting to the International Society for Heart and Lung Transplantation (Figure 2).⁶ The difference in outcomes is partially related to a more liberal HT process in Europe with higher-risk donors utilized compared with the United States. Between 2010 and 2020, the donor utilization rate

Table 1. Number of Heart Transplants and Posttransplant Outcomes Per Country

	HT PMP (2022)	1 y mortality (year)	Waitlist mortality (year)	Constraints
North America				
United States	12.4	9.2% (2020)	5% (2016–2018)	Rural-urban divide; race and sex disparities; change in practice to meet criteria (IABP)
Canada	3.98	10% (2019)	18%* (2022)	Organ shortage, high waitlist mortality, and limited access in remote northwestern provinces
Latin America and the Caribbean				
Brazil	1.68	29% (2015)	20% (2019)	Low donor utilization rate, high waitlist mortality, outdated reimbursement models, and regional disparities
Mexico	0.32	30%†‡ (1988–2011)	NA	Lack of unifying allocation criteria, constrained health budget, and regional disparities
Western Europe				
Spain	6.6	18% (2018–2020)	5% (2011)	Declining number of available organs
France	6.39	20% (2018–2020)	14% (2018–2021)	Regional disparities
United Kingdom	2.95	12%–15% (2018–2021)	3%–5% (2018–2020)	Long waitlist time, high percentage of removal from waitlist
Central and Eastern Europe				
Russian Federation	2.0§	12%–20%†‡ (2012–2019)	27%† (2010–2017)	Lack of unifying allocation criteria, rising waitlist time, regional concentration in the western part of the country, and public trust in opt-out policy
Croatia	9.27	18%† (1995–2013)	10%‖ (2022)	Ethical obligations to DCD program, social and sex-based disparities
Germany	4.24	20% (2005–2018)	10%‖ (2022)	
Middle East and North Africa				
Israel	3.6	2%–36%† (1991–2017)	NA	Religious opposition to transplant, and ethnic disparities
Saudi Arabia	1.15	19%† (1989–2012)	NA	Religious opposition to transplant, and ratio of national/foreign donors
South Asia				
India	0.17	19%† (2012–2018)	NA	High waitlist mortality, limited MCS, and geographic constraints
East Asia				
China	0.5	14%† (2008–2014)	NA	Low donation rates, long waitlist time, and data transparency
South Korea	3.26	10% (2014–2017)	NA	Aging society, and limited organ availability
Western Pacific				
Australia and New Zealand	4.48	4%–11%† (2014–2019)	NA	Limited data on the geographic distribution
Japan	0.63	5%† (2002–2009)	31%	Cultural barriers to transplant, low rates and long waitlist times

DCD indicates donation after circulatory death; HT, heart transplant; IABP, intra-aortic balloon pump; MCS, mechanical support; NA, not available; and PMP, per million population.

*Including withdrawals.

†Single center.

‡Immediate post-HT.

§Data from 2021.

||Euro transplant.

in the United States was 44%, compared with 70% in the Eurotransplant zone, and utilized donors in the Eurotransplant zone were on average a decade older (44 versus 31 years), with a greater cardiovascular risk profile.¹⁶ Data comparing post-HT care quality metrics between the United States and Europe is scarce.

Despite the high HT number and positive outcomes, challenges persist. Waitlist mortality remains high, with significant variability in survival benefit between centers across the country (mean survival difference: 10%).^{14,17} Racial and sex-based disparities are present. A United Network for Organ Sharing analysis showed that women listed as status 1 had higher waitlist mortality compared with men. Female sex was associated with a 14% and

17% increase in adjusted risk of death for candidates listed as status 1A and 1B (pre-2018 classification), respectively.¹⁸ Black patients have a lower rate of transplant compared with their White counterparts (adjusted hazard ratio for transplant 0.81, 2018–2021) and had a 28% higher risk of graft failure (2017–2022).^{19,20} Geographic disparities in access to HT present an additional challenge; rural areas have a 22% lower HT rate compared with urban areas.²¹

The 2018 allocation system changes achieved some intended goals of more accurate stratification, broader geographic sharing of organs and decreased waiting times.²² While some goals were met, challenges remain. To decrease subjectivity, the 2018 system outlined

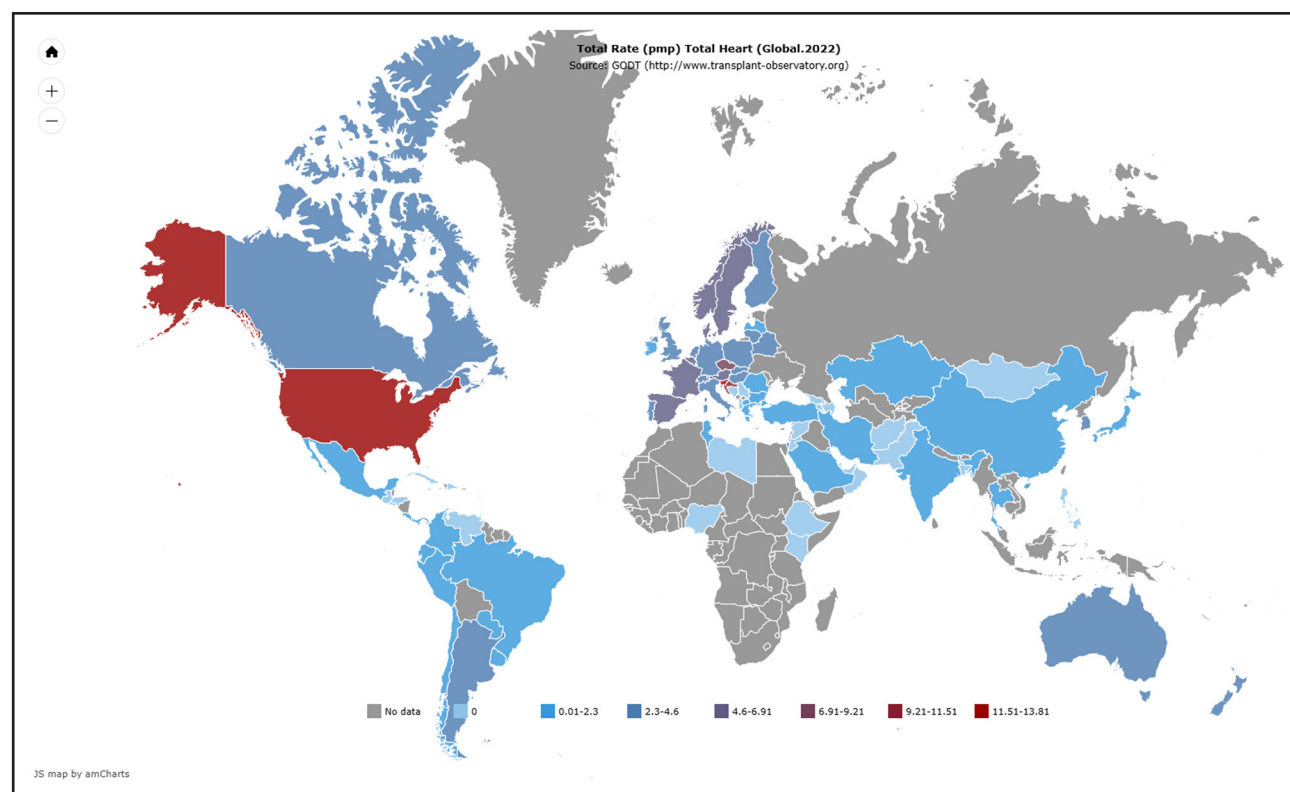


Figure 1. Global adult heart transplant activity per million population 2022.

Reproduced with permission from the Global Observatory on Donation and Transplantation (GODT). Organ Donation and Transplantation Activities 2022 Report. https://www.transplant-observatory.org/wp-content/uploads/2016/02/2022-data-global-report_VF_2.pdf

hemodynamic criteria required to initiate temporary mechanical support (MCS) or inotrope for potential HT candidates; however, it left the decision to the treating physician.¹³ Since then, the number of MCS listings has doubled with no significant change in patients' characteristics, reflecting a change in practice to warrant prioritization in listing status rather than a change in the population.²² A recent study evaluating a novel continuous allocation score found that the use of an intra-aortic balloon pump (IABP) overestimates the candidate risk at the expense of candidates listed as low priority. More than 1 in 4 candidates listed as low priority (status 3–6; Table 2) were at high risk for death using a continuous scoring system.²³ The implementation of extended geographic sharing in the 2018 system increased ischemic times.²² Though posttransplant outcomes improved for Black patients compared with historical data, the rate of transplant and waitlist outcomes may have worsened under the new system, a finding that requires further study.

Canada

Canada, a nation of 40 million, allocates 12% of its GDP to health care. Canada has a well-developed health care framework with better HF care delivery metrics compared with the United States.²⁴ The burden of HF in Canada is high with 750 000 people living with HF

and a cost that is expected to reach nearly \$3 billion per year. However, HT care in Canada lags behind the United States; in 2022, there were 3.98 HTs PMP, compared with 12.4 HTs PMP in the United States.⁵

The Canadian Cardiac Transplant Network and Canadian Organ Replacement Register publish annual reports on HT activity in Canada. The annual report provides comprehensive data on HT rates, waitlist data and outcomes. The most recent HT allocation system was revised in 2021.²⁵ The new allocation system is a 5-tier candidacy listing system, compared with 4 before 2021. The system created distinctions between temporary and durable MCS and better defined the type of MCS support required for the highest category (status 4) and excluded IABP from that category. Furthermore, the new system better defined unstable arrhythmia patients and categories for combined organ transplant.

The number of HTs in Canada in the last 2 years has declined. In 2022 there was a total of 125 HTs (excluding combination transplants) performed, a 21% decrease compared with 2020 and a 33% decrease compared with 2017.²⁶ It is postulated that the COVID-19 pandemic played a role in the decline; however, the reasons for the disproportionate decline compared with the United States are unknown, and process reviews are underway to explore reasons for the decline. Unadjusted (excluding Quebec), 1-year post-HT mortality was around 10%

Table 2. Comparison of Heart Transplant Waitlist Criteria Across 4 Regions

United States	Eurotransplant	France	South Korea
Status 1: VA-ECMO Nondischargeable, surgically implanted, nonendovascular biventricular support device MCS with life-threatening ventricular arrhythmia	International HU status (box A–C): Box A: Inotrope dependency Cardiac hypoperfusion (3/3): 1. Clinical features: NYHA ≥3. 2. Hemodynamic features: CI <2.0 L/min-m ² , SvO ₂ <50%, PCWP >15 mm Hg. 3. Signs of organ damage, at least 1: lactate >reference, bilirubin >reference, GFR <60 mL/min, Na <135 mmol/L Inotrope dependency (1/2): For least 72 h with 1. CI <2.0 L/min-m ² 2. Return of cardiac perfusion syndrome within 14 d after infusion of levosimendan.	Cardiac risk score (CRS)=1.301335×VA-ECMO+0.157691×NP decile−0.510058×ln (GFR)+0.615711×ln (bilirubin). CRS is then converted to a Cardiac Risk Index (CRI) by the following formula: CRI=min (40 [CRS+1.41]×10) to obtain an integer ranging from 0 to 40. The CRI is transformed into a score with an upper bound of 1000 instead of 40, by applying the following formula: score=1000×CRI/40.	Status 0: VA-ECMO VT/VF needs VAD VT/VF needs IABP VAD with complication External VAD
Status 2: Nondischargeable, surgically implanted, nonendovascular LVAD IABP VT/VF, mechanical support not required MCS with device malfunction/mechanical failure TAH, BiVAD, RVAD or VAD for single ventricle patients Percutaneous endovascular MCS	Box B: High urgency criteria for patients with VAD/TAH. VAD/TAH with complication (cerebral, embolic, GI bleed, RV failure, infection, mechanical failure and thrombosis). Box C: Exceptional cases Hypoperfusion on short-term MCS, re-transplantation, amyloid, HCM, CHD and uncontrollable arrhythmias.	Allocation exceptions: Durable MCS with device-related complications, as well as in candidates on uncomplicated BiVAD and TAH. Life-threatening ventricular arrhythmias. The situation is addressed by assigning score exceptions to these candidates 900 points are allocated immediately or over 3 mo to patients on durable MCS with device-related complications and to patients on uncomplicated BiVAD and TAH. Requests for exceptions to the CRS are reviewed.	Status 1: Inpatient, with at least 1 of the following: TAH, VAD, IABP, continuous IV inotropes >4 wk, continuous IV inotropes a single high-dose or multiple moderate dose >1 wk, VT/VF >3 times in 24 h, >3 ICD shocks
Status 3: Dischargeable LVAD for discretionary 30 d Multiple inotropes or single high-dose inotrope with continuous hemodynamic monitoring VA-ECMO after 7 d; percutaneous endovascular circulatory support device or IABP after 14 d Nondischargeable, surgically implanted, nonendovascular LVAD after 14 d MCS with a complication			Status 2: Continuous infusion doesn't fulfill criteria for status 1 VT/VF despite AAD, ICD shock not fulfilling criteria for status 1
Status 4: Dischargeable LVAD without discretionary 30 d Inotropes without hemodynamic monitoring Retransplant Diagnosis of 1 of the following: CHD, IHD with intractable angina, HCM, RCM and amyloidosis	T-Transplantable: The urgency status transplantable (T) is used for elective transplant candidates who are not eligible for the urgency statuses HU or ACO.		Status 3: Patients that do not meet the criteria for status 0, 1 or 2
Status 5: Dual organ	Approved combined organs (ACO): For patients in need of a multiorgan transplant, except the combination heart with kidney and heart/lung with kidney—ACO status can be requested.		Status 7: Deferred HT listing
Status 6: All remaining active candidates			

AAD indicates antiarrhythmic drugs; BiVAD, biventricular assist device; CHD, indicates congenital heart disease; CI, cardiac index; ECMO, extra corporeal membrane oxygenation; GFR, glomerular filtration rate; GI, gastrointestinal; HCM, hypertrophic cardiomyopathy; HU, high urgency; IABP, intra-aortic balloon pump; IHD, ischemic heart disease; IV, intravenous; LVAD, left ventricular assist device; MCS, mechanical circulatory support; Na, sodium; NYHA, New York Heart Association; PCWP, pulmonary capillary wedge pressure; RCM, restrictive cardiomyopathy; RV, right ventricle; RVAD, right ventricular assist device; SvO₂, mixed venous saturation; TAH, total artificial heart; VA, veno-arterial; VAD, ventricular assist device; VF, ventricular fibrillation; and VT, ventricular tachycardia.

in 2019, a number that remained steady over the past decade.²⁷ In 2022, there were 119 on the HT waitlist in Canada. Waitlist mortality in Canada is high; 22 HT candidates (18%) withdrew or died before receiving an HT in 2022.²⁶

Several factors contribute to high waitlist mortality in Canada. Organ shortage is a major issue in Canada,

with an organ donation rate of 19.3 donors PMP in 2021, compared with 42.0 donors PMP in the United States.²⁸ Newer organ procurement techniques are yet to be widely implemented for HT in Canada; in 2022, there were 0 HT from DCD.²⁶ Disparities in HT access to Indigenous peoples (First Nations, Métis or Inuit patients) and individuals in remote areas are present.²⁹

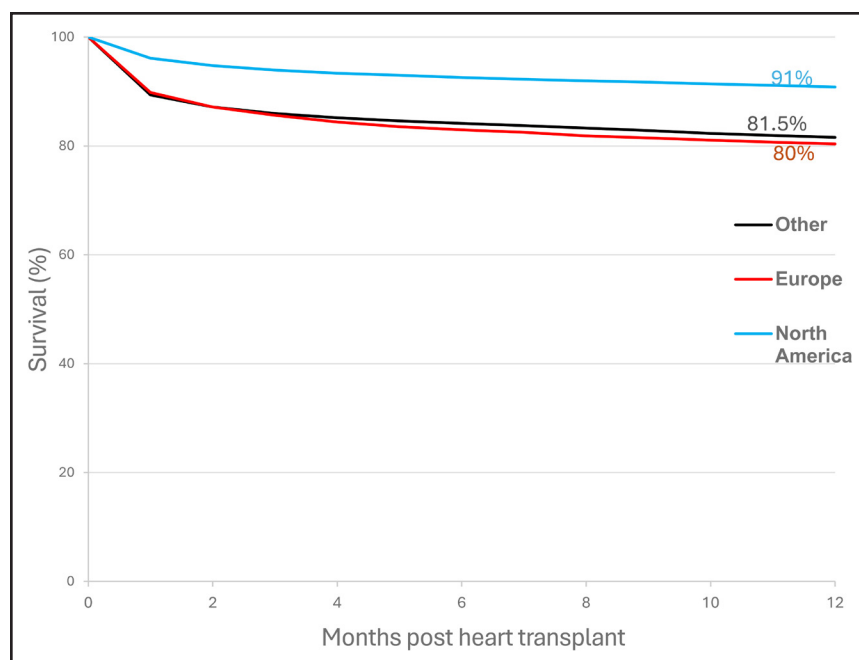


Figure 2. Kaplan-Meier plot for post-heart transplant survival by region (January 2012–June 2017).

Data reproduced from the International Thoracic Organ Transplant Registry of the International Society for Heart and Lung Transplantation: Thirty-Eighth Adult Lung Transplantation Report-2021.¹

Despite having a well-developed health care system, Canada has a relatively small number of HTs PMP. Organ shortages, limited adoption of advanced procurement methods and disparities in access for indigenous populations pose a challenge to Canada's HT landscape.

LATIN AMERICA AND THE CARIBBEAN

Over the past 30 years, substantial progress with respect to HT has been made in Latin America and the Caribbean. According to international registries in 2022, Argentina had the highest number of HT at 2.39 PMP, followed by Uruguay at 2.29 PMP, Chile at 2.07 PMP and Brazil at 1.68 PMP.⁵ HT in Latin America and the Caribbean is generally characterized by limited referrals of advanced HF patients to HT. HT is limited to a few centers that perform fewer HTs than expected, primarily due to a low donor utilization rate. However, there is growth potential for donor availability with improved logistics and enhanced donor intensive care management. Waitlist mortality remains extremely high, particularly in centers with a predominance of cardiogenic shock patients. Intra-aortic balloon is the only disposable MCS in most centers, and other MCS options are scarce or entirely absent due to economic constraints.

Brazil

Brazil, a country of 211 million people, allocates 9% of its GDP to health care, with HF carrying the second highest financial burden to the health care system.³⁰ Brazil has the highest level of public funding for transplants, and public funding covers 95% of transplants in Brazil.³¹ HT cost in Brazil is 50% lower compared with the United

States. The Brazilian Association of Organ Transplantation, established in 1986, regulates organ transplant activity.³⁰ Brazilian Association of Organ Transplantation publishes an annual report on solid organ transplant activity in Brazil. While the annual report provides comprehensive data, including waitlist mortality data and regional statistics, post-HT outcomes are not provided.³¹

In 2022, there were 1.68 HT recipients PMP in Brazil.⁵ The number of HT increased from 228 to 380 from 2012 to 2019, a 40% increase, the highest total number of HT in Latin America and the Caribbean.³¹ The number of effective organ donors rose from 12.6 PMP in 2012 to 18.1 PMP over the past decade, reflecting a positive attitude toward transplant among Brazilians. A survey showed that 70% of Brazilians are willing to donate organs in case of death.^{31,32} The Brazilian HT allocation system was last updated in 2010. Unlike the United States and Canada, there are no tiers to the allocation system beyond the highest urgency status. The listing system prioritizes patients needing MCS, on temporary inotropes, and mechanically ventilated patients. However, the lack of objective listing criteria makes the system prone to subjectivity. Similar to the United States, IABPs are given the same priority as other MCS devices.³³

HT in Brazil faces several challenges. National post-HT survival data are not available. Published literature indicates that 1-year survival post-HT in Brazil is low. An analysis of HT recipients in Brazil between 2000 and 2015 found a 1-year survival of 71%.^{34,35} Waitlist mortality is also high, 20% (116/558) in 2019. This is likely related to organ shortages; despite the public's positive attitude toward transplants, donor utilization remains well below 40% (44% in the United States and 70% in Euro transplants).³⁶ It is estimated that the number of HTs

performed in 2017 was 23% of the country's HT need (≈ 8 PMP).³⁶ National registries face limitations due to missing data, including the lack of cause of HF (70% of cases), use of posttransplant immunosuppression, causes of death and rate rejection or vasculopathy.³⁴ Although public funding availability is considered positive, compensation models remain well below the necessary margin. Recent data show that reimbursement is 60% of the actual cost.^{36,37} Finally, similar to the United States, regional inequalities exist, with most HTs occurring in the southeast and south regions, while no HTs were reported in northern Brazil in 2019.³⁸

Mexico

Mexico's HF burden is high with an estimated annual prevalence of $\approx 2\%$. Mexico allocates 5.5% of GDP to health care with HF expenditure accounting for $\approx 4\%$ of all health costs. The first Mexican statement on HF published in 2021 provides guidance on advanced HF care and patient selection for HT; however, no specific listing criteria are provided.³⁹ The National Transplant Center, established in 2000, oversees organ donation and transplant activity and publishes a detailed semi-annual report on HT rates, patient characteristics and geographic distribution.

Mexico has a low HT rate, with only 0.32 HTs PMP in 2022.⁵ HT constitutes a mere 1% of solid transplants in Mexico.^{40,41} The National Transplant Center reports a decrease in the total HTs over the past 5 years, with 26 HTs in 2021 compared with 39 in 2015.⁴² Among 57 authorized transplant centers, only 10 had HT activity, with 1 center performing more than a third of all Mexico's HTs.⁴⁰ The limited number of HTs can be attributed to Mexico's constrained health budget and organizational challenges in both governmental and hospital administrations.⁴¹ There is no national post-HT outcome data, and data are limited to single-center experience. A review of HTs conducted between 1988 and 2011 revealed an immediate post-HT mortality of 30.3%, markedly higher than International Society for Heart and Lung Transplantation-reported rates; more recent outcomes data are not available.⁴³

WESTERN EUROPE

Western Europe, with an aging population, has a high number of prevalent HF cases.² There is significant variability across countries in Western Europe in terms of HF incidence and outcomes.⁴⁴ Western European nations have diverse HT policies and allocation systems. In this section, we will briefly review 3 of the largest HT systems in Western Europe.

Spain

Spain, a country of 47 million people, has the highest life expectancy in Europe and is one of the world leaders

in policy development for solid organ transplants. The National Transplant Organization is an independent coordinating agency of the Spanish Ministry of Health and is the governing body for organ transplant activity in Spain. The Spanish Heart Transplant Registry provides an annual report with comprehensive data on HT activity in Spain. The waitlist criteria were most recently updated in 2023 and are a 3-tier system (urgency level 0, urgency level 1 and elective). The system prioritizes patients on MCS and places less emphasis on continuous inotropy.⁴⁵ Unlike the United States, IABPs are not prioritized (since 2017), and that is reflected in recipients' data; the use of IABP pretransplant has decreased from 15.4% to 2.4% between 2012 and 2021; however, the use of total ventricular support increased from 9.8% to 29%. These changes have led to more appropriate patient selection with a decrease in urgent transplants and an increase probability of elective transplants without changes in elective transplant waiting times.⁴⁶

In 2022, Spain had an HT rate of 6.6 PMP, one of the highest globally.⁵ There were 302 HTs performed in Spain in 2021, representing an 8.6% increase compared with 2020. This is in keeping with trends over the past decade.⁴⁶ One-year survival post-HT between 2018 and 2020 was 81.5%, slightly improved compared with historical data (78.3%, 2012–2014), similar to other European countries and lower compared with the United States.⁴⁶ Waitlist is low and was 5%, with a median wait time of 122 days, in 2011.⁴⁷ Outcomes are likely related to a more aggressive transplant process compared with the United States; donors are, on average, a decade older (mean age, 41.5 versus 31 years). Low waitlist mortality is likely related to a high donation and utilization rate. Spain has the highest donation rate in the world, and utilization rates have been reported to be $>80\%$ (compared with 44% in the United States).⁴⁸ Spain main challenge is the low number of available organs, driven by a decrease in cardiovascular and traffic-related deaths. Frameworks for expanding the donor pool have been described and include more use of high-risk donors and expansion of DCD. Currently, the percentage of HT from DCD donors is not provided in Spain.⁴⁸

France

France is the second-largest country in Western Europe with a population of 67.8 million in 2022 with a higher than the European average health expenditure per capita, representing 12.2% of GDP.⁴⁹ The French agency of Biomedicine (Agence de la Biomedecine), a public establishment, manages HT activity in France.⁵⁰ The CRISTAL registry (Communication, Répartition, Information, Suivi, Traçabilité, Analyses, Logistique), initiated in 1996, harbors data on HT activity in France and is maintained by the Agence de la Biomedecine.

The Agence de la Biomedecine publishes an annual report on HT activity in France. The annual report details

trends on the HT waiting list, rate of transplants and post-HT outcomes. In 2021, there were 409 HTs (6.0 PMP), a 10.5% increase compared with 2020. However, HT activity is yet to reach prepandemic levels; in 2015, there were 471 HTs in France.⁵¹ Cumulative 1-month survival post-HT between 2018 and 2020 was 88.8%, stable compared with 88.6% for HT recipients between 2013 and 2017.⁵¹ Cumulative 1-year survival post-HT between 2018 and 2020 was 80.1%, stable compared with 79.8% for HT recipients between 2013 and 2017.⁵¹

In 2018, France adopted a new HT allocation system, transitioning from a tier-based system to a score-based system. HT candidates are prioritized based on a candidate risk score (CRS). The CRS formula is derived from Veno-arterial-extracorporeal membrane oxygenation use, serum natriuretic peptide, glomerular filtration rate and bilirubin level. The French system does not grant priority to patients on inotrope infusion or LVAD patients, nor does waiting time provide any points for a candidate's waitlist time. The allocation system allows for exceptions for HT candidates whose risk is not well assessed by the CRS (Table 2).⁵² The CRS is a data-driven approach and avoids heterogeneous patient categorization. Implementation of CRS on a United States cohort outperformed the 6-tier-based system.²³ However, concerns for longer wait times in France, particularly among LVAD patients, have been raised.⁵² Patients registered for HT after the implementation of the new allocation system had longer wait times (an average of 0.5 months longer) compared with those registered prior, with slightly higher mortality (14% versus 12%).⁵¹ Assessment of the new allocation system was limited by the COVID-19 pandemic. A recent competing risk analysis found that the new system increased the incidence of HT with no change to waitlist mortality.⁵³ Regional disparities pose a challenge to HT in France, with the HT rate ranging from 9.6 PMP (Northeast, Champagne-Ardenne) to 2.7 PMP (West, Poitou-Charentes).⁵¹

The United Kingdom

Unlike other Western European countries, the United Kingdom has its own National Institute for Health and Care Excellence, guiding HF management.⁵⁴ In 2022, the United Kingdom ranked fifth in Europe for health care spending.⁴⁹ The UK National Health Services Blood and Transplant oversees organ donation and HT activities. Unique to the United Kingdom, organ allocation policy is updated annually. The policy is developed by an advisory group (the Cardiothoracic Advisory Group) and is reviewed by a transplant policy review committee. As of 2023, the organ tissue donation and transplant clinical audit risk and effectiveness group is responsible for the annual review.⁵⁵ The most recent update on HT allocation criteria was in 2016 and included a third tier to better reflect the acuity of patients (Table 2).

The National Health Services Blood and Transplant publishes an annual report on cardiothoracic transplant activity. According to the 2022–2023 report, there were 214 HTs in the United Kingdom (3.2 HTs PMP), a 16% increase compared with the year prior ($n=179$).⁵⁶ Remarkably, 26% ($n=55$) of HT were recovered from DCD donors, a 20% increase in DCD compared with a year prior. Post-HT 1-year survival rates are reported separately for donation after brain death and DCD. One-year survival between 2018 and 2021 for post-donation after brain death and DCD HT was 85% and 88%, respectively.⁵⁷ Posttransplant outcomes are closer to the United States and better than the European average. The median donor age (35 years) is younger than the European average. The United Kingdom reports a low waitlist mortality; 1-year waitlist mortality between 2018 and 2020 was 5% and 3% for nonurgent and urgent, respectively. Super-urgent HT patients had a 4% 3-month waitlist mortality. However, due to long waitlist times, a significant percentage of candidates (20% in urgent and 30% in super-urgent) were removed from the transplant list within 1 year or 3 months, respectively.⁵⁶ The overall national waitlisting time between 2015 and 2021 was 845 days (compared with 180 days for the Eurotransplant). For the same period, the waitlist median time for urgent and super-urgent was 47 and 14 days, respectively.⁵⁶

A regional variation analysis showed no difference in HT rates across England.⁵⁸ The analysis did not include other UK countries. A single-center review identified sex-based differences in characteristics; however, no systemic bias against females was noted. Male patients with blood type O experienced lower transplant rates and higher mortality, linked to a scarcity of suitable donors.⁵⁹

CENTRAL AND EASTERN EUROPE

Central and Eastern Europe are geopolitical terms that encompass a variety of nations with differing governing models and a heterogeneous population. The burden of HF in Central and Eastern Europe is high, with an age-standardized rate of 560.22 per 100 000 population, and it is one of the few regions with a rising prevalence of HF.² In this section, we review HT systems in countries that reflect the diverse nature of this region.

The Russian Federation

The Russian Federation is the largest territory in the world with a population of over 145 million. Cardiovascular disease (CVD) accounts for nearly half of the country's fatalities.⁶⁰ Russia on average spends 5% of its GDP on health care. Despite its historical significance in intrathoracic transplant, Russia's current HT activity is low.⁶¹ The Russian Transplant Society tracks HT activity and publishes an annual report on organ transplant activity in the United States.⁶² Unlike the United States, and similar to

Brazil, the report is not specific to HT and lacks outcomes data. In 2021, there were 2.0 HTs PMP in Russia, totaling 268 HTs across 17 centers.⁶² HT centers have expanded across the country, with new centers being established in remote areas. In 2007, the first HT was performed in Siberia. However, most HTs still occur in large urban areas, with the Shumakov Center in Moscow accounting for 73% of all HTs in Russia.⁶¹ In 2021, there were 4.5 PMP donors, an effective increase of $\approx 16\%$ compared with 2020. Despite this, HT waitlist times have increased 1.8 $\times$ over the past decade. Reasons behind the increase in waitlist times and average waitlist times are not provided.⁶²

Outcome data are limited to single-center experiences. A single-center analysis of 52 HT patients between 2012 and 2019 found in-hospital survival to be 80.0% to 88.5% depending on ischemic time.⁶³ A review of 151 HT recipients between 2010 and 2017 found waitlist mortality to be 27%.⁶⁴

Despite having robust institutions, the HT system in Russia faces several challenges. HT rates are much lower compared with other European nations. There is no standardized allocation system, and practices may vary from center to center. Active HT centers are highly concentrated in the western region and do not cover the country's vast territory. Public trust in the transplant system is challenged, in 1992 a law on the transplant of human organ and tissue was passed, establishing an opt out system for organ donation.⁶⁵ The public is not fully aware of the opt out policy for organ donation, and this has led to litigation, prosecution and even periodic suspension of transplant programs.⁶⁶

Eurotransplant: A Multinational Organization for Organ Allocation

Eurotransplant is a nonprofit, multinational organization that facilitates organ allocation and transplant across its member states. Founded in 1967, it currently serves 8 European countries: Austria, Belgium, Croatia, Germany, Hungary, Luxembourg, the Netherlands and Slovenia. Together, these nations represent a population of ≈ 139 million people, making Eurotransplant a critical hub for organ sharing in Europe. The primary mission of the Eurotransplant is to ensure equitable and efficient organ allocation among its member countries. This is achieved through a centralized database that matches donor organs to suitable recipients based on medical urgency, compatibility and logistical considerations. The Eurotransplant operates a democratically structured framework, allowing member states to contribute to policymaking and organ-sharing protocols. In 2023, the Eurotransplant coordinated 645 HTs; at the end of 2023, 1044 patients were actively waiting for an HT, and 1123 new patients were added to the waiting list during the year. The median waiting time for HT was 125 days, with a waitlist mortality rate of 9%.⁶⁷ Of the total HTs performed, 261 (40%)

were performed under the high urgency designation, highlighting the importance of this system in saving lives. Donation rates across the Eurotransplant zone vary, with Croatia leading with 27.7 deceased donors PMP, while other countries like the Netherlands and Germany report rates of 15.5 and 10.8 donors PMP, respectively.⁶⁷ The reason behind high donation rates in Croatia is the long-term strong governmental support and legislation that encourage donation. Croatia, like Russia, has an opt out policy adopted in 1988; however, it was coupled with wide and continuous messaging campaigns leading to public buy-in. The disparities emphasize the need for continued collaboration and efforts to increase organ donation rates across member states. HT candidates are categorized into 3 urgency criteria: high urgency, approved combined organ and transplant for elective HT (Table 2).⁶⁸ To obtain high urgency status, centers must submit comprehensive medical data to an independent audit team for review. If no suitable donor is available nationally, high urgency cases can access organs from the broader Eurotransplant network. The Eurotransplant's framework enables smaller countries, such as Luxembourg and Slovenia, to access a larger donor pool, fostering equitable sharing. The centralized system also supports data collection and research, driving advancements in transplant.

Challenges remain, including longer average wait times of 6 months compared with regions like the United States. Variability in donor practices, such as the adoption of DCD, persists. Countries like Croatia and Germany do not perform DCD due to ethical considerations, while others have adopted it to expand the donor pool. Outcome data is reported by individual states and is limited in smaller countries. For example, English literature searches yield limited results on Croatian post-HT outcomes. Single-center reports of 85 recipients between 1995 and 2022 showed an in-hospital mortality of 10% to 12% and a 1-year survival rate of 82% to 84%.^{6,69,70}

MIDDLE EAST AND NORTH AFRICA

The Middle East and North Africa region is a large and diverse region, with 6% of the world's population, roughly equivalent to the European Union. The Middle East and North Africa is one of the few regions experiencing a rise in prevalent HF cases.² This increase is likely due to the high burden of cardiac risk factors in the region. High and middle-income countries within the Middle East and North Africa have made significant strides in improving health outcomes over the past decade, paving the way for the development of HT systems.⁷¹

Israel

Israel has one of the highest health metrics rankings and spends nearly 8% of GDP on health care. However, the nation's transplant system was limited until 2008 when

the first Israeli organ transplant and brain-death-related law was passed.⁷² The National Transplant Center (ADI) is responsible for overseeing transplant activity. Data for all patients awaiting transplants in Israel is stored by ADI, but details are not publicly available.

According to GODET, Israel had 3.6 HT recipients PMP in 2022.⁵ Single-center data of 70 HTs between 2010 and 2017 showed in-hospital mortality of 10%.⁷³ According to data published by ADI, there was a record high of 32 HTs in 2022, a rise from 24 HTs in 2021 and 9 HTs in 2014.⁷⁴ Israel's HT allocation system is tiered based on medical urgency (status 1A, 1B and 2), with a unique point system that prioritizes patients based on the 2008 organ donation law. This system gives higher priority to patients registered as organ donors for at least 3 years before listing or to those with a first-degree relative registered as a donor.⁷⁵

Despite advancements since the implementation of new laws, challenges persist, particularly in the ultra-orthodox community where many do not accept brain death as the end of life for religious reasons. Additionally, disparities in HT outcomes based on ethnicity have been observed. A study of 254 HT recipients from 1991 to 2017 revealed that Arab patients had more than a 4-fold increase in posttransplant mortality (5-year mortality: 36% versus 1.7%) compared with their Jewish counterparts.⁷⁶

Saudi Arabia

Saudi Arabia (SA) is the largest country in the Middle East with a rapidly growing economy and expanding population that spends nearly 6% of GDP on health care. The country faces a significant burden of CVD and HF due to the high prevalence of cardiac risk factors.^{2,77} King Faisal Specialist Hospital is the country's major transplant center and averages around 30 HTs a year.⁷⁸ There is no publicly available registry for outcomes of patients with HT in SA, and outcome data are limited to single-center experiences.⁷⁹

In 2022, 1.15 HTs PMP were performed in SA.⁵ Available data from King Faisal Specialist Hospital between 1989 and 2012, including 164 HT patients, showed post-HT survival of 88% and 81% at 30 days and 1 year, respectively.⁸⁰ This is comparable to Eurotransplant outcomes.

SA HT system faces organ shortage and cultural challenges. Most donors are non-Saudi expatriates, with only 1% to 6% of donors being Saudi nationals. A survey of 1250 Saudis revealed that while 91% supported organ transplant in principle, 42% opposed undergoing HT for religious reasons and only 43% were willing to donate their hearts. Additionally, 31% expressed concerns about potentially receiving less medical effort to save their lives if registered as organ donors.⁸¹

SUB-SAHARAN AND SOUTH AFRICA

Sub-Saharan Africa accounts for 80% of the African continent, with 1.1 billion people across 47 countries.⁸² Since performing the world's first HT in 1967 at Groote Schuur Hospital in Cape Town, limited advancements have been made in making HT widely available. Of the region's 62 transplant centers in 7 countries, only South Africa performs HTs, with 42 HTs in 2018. This accounts for 0.09 HT PMP for the region and 0.73 HT PMP for South Africa.⁸³ To our knowledge, there are no published outcomes data for HTs in Sub-Saharan Africa.

The 2009 Abuja Declaration was a product of efforts by the World Health Organization aimed at promoting legislation on donation and transplant in the region. Nevertheless, many transplant systems remain fragmented, with limited infrastructure and a paucity of expertise and institutional support. In the few countries that have transplant centers, the individual programs for donation and transplant of organs and tissues are not consolidated.⁸³ Furthermore, posttransplant is limited as adequate immunosuppression is often unavailable, and if available, it is cost-prohibitive for patients who pay out of pocket.⁸² Adequate histological evaluation of biopsy specimens is limited, making prompt management of rejections unavailable.

Solid organ transplant systems depend on robust and adequate finances. While health expenditure in Western countries exceeds \$1800 per capita, it is only \$51.60 in Africa.⁸² The rural-urban divide is more pronounced in Sub-Saharan Africa as transplant centers are located in urban cities or state capitals that are inaccessible to rural dwellers.⁸² Local practices preclude desecration of the body of the deceased and are a reported factor prohibiting family members from donating body parts of their deceased relatives.⁸²

SOUTH ASIA (INDIA)

South Asia is dominated by the Indian subcontinent. The first HT was attempted in India in 1968 in Mumbai.⁸⁴ It was not until 1994, with the passage of the Transplantation of Human Organs Act, that deceased organ donation was formalized.⁸⁴ The National Organ and Tissue Transplant Organization, established in 2014, manages HT activity in India.⁸⁵ Candidates are listed based on a 3-tier system based on illness: priority 1 (emergency), priority 2 (semi-emergency) and priority 3 (elective).⁸⁵

In 2022, there were only 0.17 HTs PMP in India.⁵ The National Organ and Tissue Transplant Organization annual report in 2022 recorded 250 HTs in India, a significant increase from 151 HTs in 2021 and comparable 241 HTs in 2018, prepandemic.⁸⁶ The report lacks data on geographic statistics, waitlist outcomes and post-HT outcomes.

Outcome data are limited to single-center reports. A study from 2012 to 2018, including 168 HT patients, reported a 1-year survival of 81%.⁸⁷ Waitlist mortality is reportedly high, and there is a paucity of MCS support options in India.⁸⁶ India's HT system faces logistical and economic challenges. Several cities where HT occurs lack functioning airports, and helicopters, mostly unavailable, cannot operate at night, leading to the loss of potential organs.⁸⁶ India has 78 transplant centers that perform HTs. In 2019, a donor's heart was transported over 200 miles to Mumbai in 80 minutes and successfully transplanted, a significant milestone for the country.⁸⁷ HTs are not government-funded, creating wealth-based disparities in access. With costs starting from 30 000 dollars, the wealthy and transplant tourists have better access to HT in a country where the average per capita annual income is estimated at \$1800 US dollars.^{86,87}

EAST ASIA

East Asia, home to 20% of the world's population, has <5% of all of HT worldwide.²⁶ There are notable differences in HT access and post-HT care within the region. In this section, we aim to contrast the HT systems of 2 developed neighboring countries in East Asia.

China

China, the world's second most populous country and the second-largest economy, spends about 5% of its GDP on health care. With an aging population, China faces a high burden of CVD; in 2019, the country had the third highest prevalence of HF in the world (age-standardized rate of 1032 per 100 000 population).² The country began reforming its transplant medicine in 2005, establishing a legal framework for organ transplant.⁸⁸ The China Heart Transplant Registry and first transplant guidelines were introduced in 2010, but brain death criteria were not established until 2013.⁸⁸ Organ allocation is through the China Organ Transplant Response System, which prioritizes patients based on a score calculated from clinical data.⁸⁹ Annual, national HT data are not publicly available and are sourced from individual institutions and occasional health ministry reports.

According to GODT, China had only 0.5 HTs PMP in 2022.⁵ A 2014 report from the Chinese organ transplant network noted 1291 registered HT candidates by mid-2013, with 232 transplants performed that year. A study from the Union Hospital in Wuhan between 2008 and 2014, including 203 HTs, showed a postoperative survival rate of 90.79% and a 1-year survival of 86.20%, comparable to US outcomes.⁹⁰

Despite good outcomes, China struggles with extreme organ shortages. Estimates in 2013 indicated 200 000 to 300 000 were people awaiting a solid organ transplant,

with only 2271 organs donated.⁹¹ The number of HT recipients in China PMP is <5% of that in the United States (0.51 PMP versus 12.4 PMP).^{5,7} Organ shortage is fueled by the public's negative attitude toward transplants and strict policies for organ donation. To be listed as a potential donor, one needs written consent from all family members. After donor death, one relative objection can cancel the donation.⁹¹

China has faced criticism for using organs from executed prisoners. By the end of 2014, China declared this practice illegal and announced that it had achieved the goal of ceasing these practices, although allegations of continued organ procurement, especially among minorities, persist, which the Chinese government denies.^{90,92} Another criticism is the falsification of data. A forensic study of data by the China Organ Transplant Response System on deceased organ donation suggests systematic falsification and manipulation of data sets, and the claims are denied by the Chinese government.⁹³

South Korea

South Korea, Asia's fourth-largest economy and one of the healthiest nations in the world, is facing a rapidly aging population and one of the world's lowest birth rates, with rising concerns about an inevitable increase in CVD burden.⁹⁴ In contrast to China, South Korea has a higher percentage of GDP spent on health care (~10%) and a more robust organ transplant system. The country's first HT was performed in 1992, and in 1999, the Organ Transplant Act was enacted, legalizing donation after brain death organ transplant.⁹⁵ The Korean Network for Organ Sharing oversees HT activity,⁹⁵ while the Korean Organ Transplant Registry, established in 2013, collects transplant data in South Korea.⁹⁶ The Korean Network for Organ Sharing prioritizes HT allocation based on a tier-level system with 5 tiers (Table 2).⁹⁶

South Korea leads Asia in HT numbers, with 3.26 HTs PMP in 2022.⁵ Korean Organ Transplant Registry's 2019 report listed 106 HTs in 2016 and 110 HTs in 2017. A notable 33% of recipients in 2017 were on extracorporeal membrane oxygenation before transplant, compared with just 2% in the United States.^{97,98} Despite the high use of MCS, South Korea reports excellent post-HT outcomes, with a cumulative 1-year survival post-HT of 90% between 2014 and 2017.⁹⁷

WESTERN PACIFIC

The Western Pacific is a geographic region that encompasses countries as large as Australia and as small as Niue, home to 100 people. There are substantial differences in population characteristics and health outcomes in the region. In this section, we review the 2 largest HT systems in the region.

Australia and New Zealand

Australasia is one of the world's largest geographic regions. The region has one of the highest life expectancies and experienced the largest (35%) decrease in HF burden over the last 2 decades.² The Transplantation Society of Australia and New Zealand oversees transplant activity in Australia and New Zealand, with the latest guidelines published in October 2022.^{99,100} The Australia and New Zealand Organ Donation Registry records organ donation activities and publishes monthly statistics and an annual report on HT activity. HT candidates are classified as urgent or nonurgent, and allocation is left at the discretion of the local transplant unit director.¹⁰⁰

In 2022, according to GODT, Australia had 4.5 HT recipients PMP, while New Zealand had 3.1 HT recipients PMP.⁵ The Australia and New Zealand Organ Donation Registry data show an increase in HTs, from 98 in 2017 to 112 in 2021, with a decline from a peak of 148 in 2020. The donation rate is low in Australia (7.7 PMP); however, this is offset by the excellent utilization rate of explanted hearts, which reached 95.7% in 2021.¹⁰¹ Outcome data are not reported nationally and is limited to single-center reports. A study done at St. Vincent's Hospital between 2014 and 2019, including 32 DCD HT and 177 donations after brain death, showed 1-year survival of 96% and 89%, respectively.¹⁰² St. Vincent's Hospital in Sydney is one of the world's leading centers for DCD with 74 DCD HT since 2013.¹⁰³

Japan

Japan is one of the world leaders in several health metrics, and health expenditures are higher than the world's average (11% of GDP). Japan has seen a 22% decrease in HF burden over the last 2 decades.² However, the development of transplant medicine in Japan has faced historical and cultural challenges. The country's first HT from a brain-dead donor in 1968 resulted in the recipient's death and the subsequent double murder charge against the surgeon, Juro Wada.¹⁰⁴ It was not until 1997 that legislation regarding organ donation and transplant from brain-dead patients became permissive, allowing HT to commence in Japan in 1999.¹⁰⁵ A revision to this law in 2010 attempted to align Japan with global standards.¹⁰⁵

The Japan Organ Transplantation Network (JOTN), a public association, oversees organ allocation in Japan.¹⁰⁶ JOTN publishes monthly statistics on HT activity in Japan. According to JOTN, there were 59 HTs performed in 2021, representing 0.6 HTs PMP.^{5,107} This figure shows an increase from 28 HTs in 2012 but remains below the pre-COVID-19 numbers of 84 HTs in 2019.¹⁰⁷ Japan's post-HT outcomes are superior to international averages, with a reported national 5-year survival rate of 90.5%, according to the 2021 JOTN report. Single centers

reported 1-year survival up to 95% between 2002 and 2009.^{105,107}

However, Japan has one of the longest waitlist times for HT in the world. As reported by JOTN, 25% of candidates on the HT had a waitlist time of >5 years, with the highest status candidates facing a waitlist time average of 1625 days. This corresponds to a high waitlist mortality, 31%.¹⁰⁵ Historical incidents shaping public perception, religious beliefs and cultural definitions of altruism are behind the low number of organ donations in Japan.¹⁰⁴

THE WAY FORWARD

In the 50 years since the first HT, remarkable advancements have been achieved. Pharmacological innovations have drastically changed post-HT outcomes. Donor pools have expanded due to the inclusion of historically marginal donors, better procurement techniques, and improved public awareness. However, significant challenges remain.

HT is often viewed as a public health luxury, particularly in LMIC, where it is generally considered a low-priority treatment. Constrained health care systems rightly try to prioritize access to lower-cost, higher-value therapies.¹⁰⁸ Yet, for many patients who reach end-stage HF, transplant remains the only available therapy. When applied judiciously in high-quality systems, HT can provide many quality-adjusted years of life. As such, the high cost can be offset by significant benefit, providing reasonable value for some countries. Our review highlights the close correlation between economic development and HT access (Figure 3). For instance, the GDP per capita for India is 3% of that in the United States, which performs ≈100 times more HT PMP compared with India. Nearly one-fifth of the world's population in Sub-Saharan Africa has little to no access to this therapy.

Organ shortage remains a global issue, and efforts to expand the donation pool are fraught with ethical and moral considerations. These efforts also require strong governance to regulate this complex process. Strategies like the opt out policies have demonstrated significant success when implemented thoughtfully and accompanied by effective public campaigns, as seen in Croatia. Conversely, the absence of public engagement can have detrimental effects, potentially halting a country's HT process, as occurred in Russia.

There is a notable disparity in HT access within individual countries, along geographic lines, with a rural-urban divide. These disparities extend to demographic factors such as sex, race and ethnicity. This is most studied in the United States. Countries where the majority of HTs are performed at private centers are especially susceptible to wealth disparities, and transplant tourism. In SA, 94% of donors are foreign nationals, and it

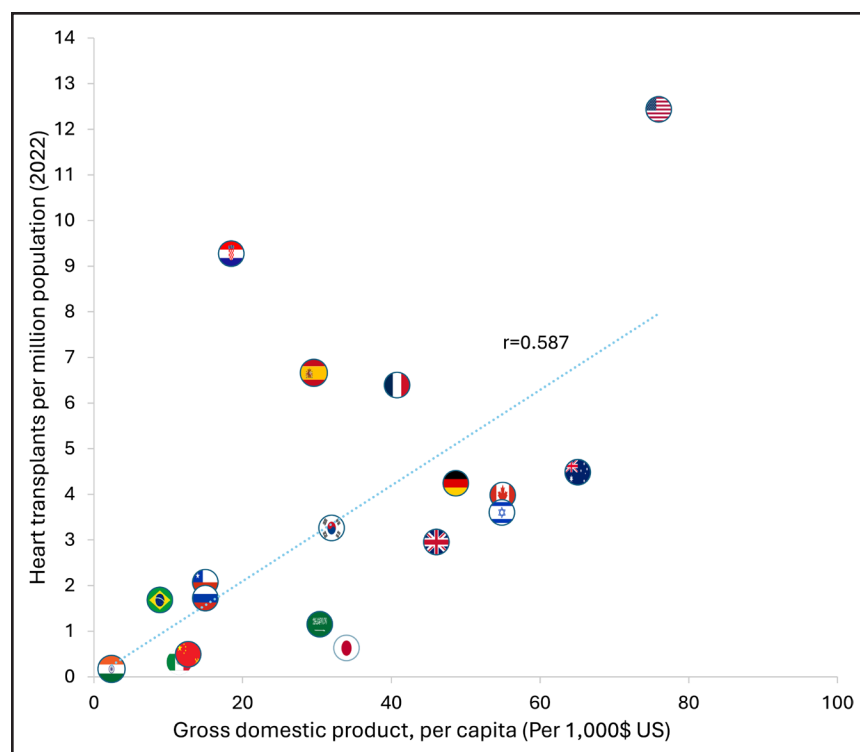


Figure 3. Correlation between economic growth and heart transplant rate.

In our review, heart transplant rates correlated with economic growth, as measured by gross domestic product per capita, with few outlier countries, such as Croatia. Heart transplant data from Global Observatory on Donation and Transplantation Russian Federation data from the International Registry in Organ Donation and Transplantation. Gross domestic product data from the World Bank DataBank.

is unclear if these individuals receive the same level of access to transplant.

There is also a lack of systematic data collection on HT recipients in many countries, both high and LMIC. In many cases, data on HT procedures are not always publicly available, and its accuracy is uncertain, as seen in China. Societies like the International Society for Heart and Lung Transplantation and World Health Organization GODT have established a standard minimum for data collection on HT procedures; however, this does not encompass all HT centers worldwide. Providing incentives to centers could help maintain the accuracy and availability of these data sets. Transparency is a core value of transplants and should be maintained by international, national and regional authorities.

Strengthening and expanding collaboration among HT systems in neighboring countries can help achieve broader access to HT worldwide. The Eurotransplant has been a success story, and efforts to replicate this model should be explored, particularly in regions with ample resources but low organ donation rates, such as the Gulf region. Additionally, a global effort to increase public buy-in and increase trust in organ donation is crucial to combat the global organ shortage.

There are ongoing efforts to strengthen legal frameworks around organ donation and transplant. The Declaration of Istanbul on Organ Trafficking and Transplant Tourism, established in 2008, was a consensus document created to curb transplant tourism and establish legal and professional frameworks within countries for organ donation and transplant activities.¹⁰⁹ The Abuja Declaration in 2009, under the World Health Organization's guidance,

aimed to promote national responsibilities in organ donation and transplant in the African region.¹¹⁰ In 2018, a new Declaration of Istanbul edition was released, emphasizing 11 principles, including equal access to transplant services, the need for legislation aligned with international standards and financial neutrality in organ donation.¹¹¹ Despite significant progress in HT activity globally, some of the core Declaration of Istanbul principles are far from reach. With shared knowledge, targeted infrastructure, public education and thoughtful application, HT could be a valuable medical treatment for a much larger portion of the world's population afflicted with end-stage CVD.

LIMITATIONS

This was not a systematic review, and despite relying on national registries when available, some evidence may have been missed. Second, our attempt to highlight HT processes in several countries across different regions may have limited the depth of comparisons. Additionally, several countries with well-established HT programs were not included to avoid an overly broad scope. Finally, while we attempted to expand the focus on LMIC, this was often limited by the lack of available data from LMIC.

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REFERENCES

- Benjamin EJ, Muntner P, Alonso A, Bittencourt MS, Callaway CW, Carson AP, Chamberlain AM, Chang AR, Cheng S, Das SR, et al; American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Subcommittee. Heart disease and stroke statistics-2019 update: a report from the American Heart Association. *Circulation*. 2019;139:e56–e528. doi: 10.1161/CIR.0000000000000659
- Yan T, Zhu S, Yin X, Xie C, Xue J, Zhu M, Weng F, Zhu S, Xiang B, Zhou X, et al. Burden, trends, and inequalities of heart failure globally, 1990 to 2019: a secondary analysis based on the Global Burden of Disease 2019 Study. *J Am Heart Assoc*. 2023;12:e027852. doi: 10.1161/JAHA.122.027852
- Severino P, Mather PJ, Pucci M, D'Amato A, Mariani MV, Infusino F, Birtolo LI, Maestrini V, Mancone M, Fedele F. Advanced heart failure and end-stage heart failure: does a difference exist. *Diagnostics (Basel)*. 2019;9:170. doi: 10.3390/diagnostics9040170
- Khush KK, Cherikh WS, Chambers DC, Harhay MO, Hayes D Jr, Hsieh E, Meiser B, Potena L, Robinson A, Rossano JW, et al; International Society for Heart and Lung Transplantation. The International Thoracic Organ Transplant Registry of the International Society for Heart and Lung Transplantation: thirty-sixth adult heart transplantation report - 2019; focus theme: donor and recipient size match. *J Heart Lung Transplant*. 2019;38:1056–1066. doi: 10.1016/j.healun.2019.08.004
- (GODT) GOoDaT. Organ donation and transplantation activities 2022 report. 2022. https://www.transplant-observatory.org/wp-content/uploads/2016/02/2022-data-global-report_VF_2.pdf.
- Chambers DC, Perch M, Zuckermann A, Cherikh WS, Harhay MO, Hayes D Jr, Hsieh E, Khush KK, Potena L, Sadavarte A, et al; International Society for Heart and Lung Transplantation. The International Thoracic Organ Transplant Registry of the International Society for Heart and Lung Transplantation: thirty-eighth adult lung transplantation report - 2021; focus on recipient characteristics. *J Heart Lung Transplant*. 2021;40:1060–1072. doi: 10.1016/j.healun.2021.07.021
- International Registry of Organ Donation and Transplantation. www.irodat.org
- Grant MJ, Booth A. A typology of reviews: an analysis of 14 review types and associated methodologies. *Health Info Libr J*. 2009;26:91–108. doi: 10.1111/j.1471-1842.2009.00848.x
- GBD 2017 Diet Collaborators. Health effects of dietary risks in 195 countries, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet*. 2019;393:1958–1972. doi: 10.1016/s0140-6736(19)30041-8
- Bozkurt B, Ahmad T, Alexander KM, Baker WL, Bosak K, Breathett K, Fonarow GC, Heidenreich P, Ho JE, Hsieh E, et al; Writing Committee Members. Heart failure epidemiology and outcomes statistics: a report of the Heart Failure Society of America. *J Card Fail*. 2023;29:1412–1451. doi: 10.1016/j.cardfail.2023.07.006
- Silbergleit A, Norman E. Shumway and the early heart transplants. *Tex Heart Inst J*. 2006;33:274–275; author reply 275.
- Gordon EJ, Gill JS. US transplant policy should strengthen bridges, not build walls. *Am J Transplant*. 2016;16:1645–1646. doi: 10.1111/ajt.13713
- Shore S, Golbus JR, Aaronson KD, Nallamothu BK. Changes in the United States adult heart allocation policy. *Circ Cardiovasc Qual Outcomes*. 2020;13:e005795. doi: 10.1161/CIRCOUTCOMES.119.005795
- Colvin MM, Smith JM, Ahn YS, Handarova DK, Martinez AC, Lindblad KA, Israni AK, Snyder JJ. OPTN/SRTR 2022 annual data report: heart. *Am J Transplant*. 2024;24:S305–S393. doi: 10.1016/j.ajt.2024.01.016
- Phillips KG, Ranganath NK, Malas J, Lonze BE, Gidea CG, Smith DE, Kon ZN, Reyentovich A, Moazami N. Impact of the opioid epidemic on heart transplantation: donor characteristics and organ discard. *Ann Thorac Surg*. 2019;108:1133–1139. doi: 10.1016/j.athoracsur.2019.03.076
- Wayda B, Angleitner P, Smits JM, van Kins A, Berchtold-Herz M, De Pauw M, Erasmus ME, Gummert J, Hartysanzky I, Knezevic I, et al. Disparities in donor heart acceptance between the USA and Europe: clinical implications. *Eur Heart J*. 2023;44:4665–4674. doi: 10.1093/eurheartj/ehad684
- Sandhu AT, Woo YJ, Khush KK. Optimizing the use of heart transplant in the United States. *JAMA*. 2019;322:1772–1774. doi: 10.1001/jama.2019.16002
- Hsieh EM, Blackstone EH, Thuita L, McNamara DM, Rogers JG, Ishwaran H, Schold JD. Sex differences in mortality based on united network for organ sharing status while awaiting heart transplantation. *Circ Heart Fail*. 2017;10:e003635. doi: 10.1161/CIRCHEARTFAILURE.116.003635
- Cogswell R, Masotti M, Morris AA, Hart A, Murray T, Yancy C. Assessment of U.S. heart transplantation equity as a function of race: observational analyses of the OPTN database. *Lancet Reg Health Am*. 2022;13:100290. doi: 10.1016/j.lana.2022.100290
- Hamilton S, Wayda B, Weng Y, Zhang S, Khush K. Abstract 4144521: racial disparities in heart transplantation: long term graft survival and non-mortality outcomes. *Circulation*. 2024;150:A4144521–A4144521. doi: 10.1161/circ.150.suppl_1.4144521
- Axelrod DA, Guidinger MK, Finlayson S, Schaubel DE, Goodman DC, Chobanian M, Merion RM. Rates of solid-organ wait-listing, transplantation, and survival among residents of rural and urban areas. *JAMA*. 2008;299:202–207. doi: 10.1001/jama.2007.50
- Maitra NS, Dugger SJ, Balachandran IC, Civitello AB, Khazanie P, Rogers JG. Impact of the 2018 UNOS heart transplant policy changes on patient outcomes. *JACC Heart Fail*. 2023;11:491–503. doi: 10.1016/j.jchf.2023.01.009
- Zhang KC, Narang N, Jasseron C, Dorent R, Lazenby KA, Belkin MN, Grinstein J, Mayampurath A, Churpek MM, Khush KK, et al. Development and validation of a risk score predicting death without transplant in adult heart transplant candidates. *JAMA*. 2024;331:500–509. doi: 10.1001/jama.2023.27029
- Fuery MA, Chouairi F, Januzzi JL, Moe GW, Caraballo C, McCullough M, Miller PE, Reinhardt SW, Clark K, Oseran A, et al. Inter-country differences in guideline-directed medical therapy and outcomes among patients with heart failure. *JACC Heart Fail*. 2021;9:497–505. doi: 10.1016/j.jchf.2021.02.011
- Network CCT. Heart status listing revision. 2021. https://ccs.ca/app/uploads/2022/01/StatusListingRevision_November9_2021.pdf
- Information ClfH. *Summary Statistics on Organ Transplants, Wait-Lists and Donors*. 2022. CIHI. 2023. <https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.cihi.ca%2Fsites%2Fdefault%2Ffiles%2Fdocument%2Fcorr-transplant-wait-list-donor-stats-2022-data-tables-en.xlsx&wdOrigin=BROWSELINK>
- Information. ClfH. *Treatment of End-Stage Organ Failure in Canada, Canadian Organ Replacement Register, 2011 to 2020: Extra-Renal Transplants — Data Tables*. CIHI. 2023. <https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.cihi.ca%2Fsites%2Fdefault%2Ffiles%2Fdocument%2Fextra-renal-transplants-2011-2020-data-tables-en.xlsx&wdOrigin=BROWSELINK>
- Moulton D. Heart transplantation in crisis in Canada. *CMAJ*. 2002;166:492.
- Tait CL. Challenges facing Indigenous transplant patients living in Canada: exploring equity and utility in organ transplantation decision-making. *Int J Circumpolar Health*. 2022;81:2040773. doi: 10.1080/22423982.2022.2040773
- Cristelli MP, Ferreira GF, Medina-Pestana J. Global perspective on kidney transplantation: Brazil. *Kidney360*. 2021;2:2016–2018. doi: 10.34067/KID.0003612021
- Registry BT. Organ transplantation in Brazil. 2019. <https://site.abto.org.br/en/publicacao/rbt-2019-2/>
- Bittencourt PL, Codes L, Hyppolito EB, Cesar HF, Moura-Neto JA, Gomes Ferraz ML. Attitudes of the Brazilian population toward organ donation. *Kidney Int Rep*. 2022;7:2737–2740. doi: 10.1016/j.ekir.2022.09.009
- Bacal F, Neto JD, Fiorelli AI, Mejia J, Marcondes-Braga FG, Mangini S, Oliveira JL Jr, de Almeida DR, Azeka E, Dinkhuysen JJ, et al; Sociedade Brasileira de Cardiologia. [II Brazilian guidelines for cardiac transplantation]. *Arq Bras Cardiol*. 2010;94:e16–e76.
- Avila MS, Belfort DSP. Progress in heart transplantation in Brazil: is it time to build a national database? *Arq Bras Cardiol*. 2021;116:754–755. doi: 10.36660/abc.20210104

35. Mangini S, Alves BR, Silvestre OM, Pires PV, Pires LJT, Curiati MNC, Bacal F. Heart transplantation: review. *Einstein (Sao Paulo)*. 2015;13:310–318. doi: 10.1590/S1679-45082015RW3154
36. Pêgo-Fernandes PM, Pestana JOM, Garcia VD. Transplants in Brazil: where are we? *Clinics (Sao Paulo)*. 2019;74:e832. doi: 10.6061/clinics/2019/e832
37. Goldraich LA, Neyeloff JL, da Silva Etges APB, Zeilmann LG, Hastenteufel LT, Ghisleni EC, Clausell N. Heart transplantation cost composition in Brazil: a patient-level microcosting analysis and comparison with international data. *J Card Fail*. 2018;24:860–863. doi: 10.1016/j.cardfail.2018.10.011
38. Soares L, Brito ES, Magedanz L, Franca FA, Araujo WN, Galato D. Solid organ transplantation in Brazil: a descriptive study of distribution and access inequalities across the Brazilian territory, 2001–2017. *Epidemiol Serv Saude*. 2020;29:e2018512. doi: 10.5123/S1679-49742020000100014
39. Magaña SJACL, Cigarroa LJÁ, Chávez MA, Rayo CJ, Galván OH, Aguilera MLF, Ivey MJB, Rascón SR, Mendez MGF, Mendoza ZGH, et al. First Mexican statement in heart failure. *Cardiovasc Metab Sci*. 2021;32:8–85. doi: 10.35366/98870
40. Arguero-Sanchez R, Olivares-Duran EM, Sanchez-Ramirez O. Heart transplantation in Mexico. 2006–2019 productivity analysis and Pan American and Iberian comparison. *Gac Med Mex*. 2022;158:90–97. doi: 10.24875/GMM.M22000647
41. Rangel-Ugarte PM, Ramirez-Castañeda S, Mendoza-Lopez AC, Gonzalez-Hermosillo LM, Flores-Calderon O, Salazar-Hernandez J, Ramirez-Castañeda A, Roldan-Valadez E. Evolution and current circumstances of heart transplants: global and Mexican perspective. *Curr Probl Cardiol*. 2022;47:101316. doi: 10.1016/j.cpcardiol.2022.101316
42. Transplantes CND. Informative statistical bulletin national transplant center. 2021. <https://www.gob.mx/cenatra/documentos/boletin-estadistico-informativo>
43. Careaga-Reyna G, Zetina-Tun H, Lezama-Urtecho CA. [Heart transplant program at the UMAE, General Hospital Dr. Gaudencio González Garza, of the La Raza National Medical Center]. *Rev Invest Clin*. 2011;63:85–90.
44. Rosano GMC, Seferovic P, Savarese G, Spoleitini I, Lopatin Y, Gustafsson F, Bayes-Genis A, Jaarsma T, Abdelhamid M, Miqueo AG, et al. Impact analysis of heart failure across European countries: an ESC-HFA position paper. *ESC Heart Fail*. 2022;9:2767–2778. doi: 10.1002/ehf2.14076
45. Trasplantes OND. Trasplante Cardíaco Criterios De distribución. 2023. <https://www.ontes.es/wp-content/uploads/2023/06/Criterios-Distribucion-Corazon-2023.pdf>
46. González-Vilchez F, Gómez-Bueno M, Almenar-Bonet L, Crespo-Leiro MG, Arizón del Prado JM, Blázquez-Bermejo Z, Delgado-Jiménez JF, de Antonio-Ferrer M, Sobrino-Márquez JM, García-Romero E; en representación de los Equipos Españoles de Trasplante Cardíaco. Spanish heart transplant registry. 33rd official report of the Heart failure Association of the Spanish Society of Cardiology. *Rev Esp Cardiol (English ed.)*. 2022;75:923–932. doi: 10.1016/j.rec.2022.06.011
47. Almenar L, Segovia J, Crespo-Leiro MG, Palomo J, Arizón JM, González-Vilchez F, Delgado J; Spanish Heart Transplantation Teams. Spanish Registry on Heart Transplantation. 23rd official report of the Spanish Society of Cardiology Working Group on Heart Failure and Heart Transplantation (1984–2011). *Rev Esp Cardiol (English ed.)*. 2012;65:1030–1038. doi: 10.1016/j.rec.2012.07.012
48. Matesanz R, Domínguez-Gil B, Coll E, Mahillo B, Marazuela R. How Spain reached 40 deceased organ donors per million population. *Am J Transplant*. 2017;17:1447–1454. doi: 10.1111/ajt.14104
49. OECD/European Union. Health at a Glance: Europe 2022: state of health in the EU cycle. 2022. doi: 10.1787/507433b0-en
50. Dorent R, Cantrelle C, Jasseron C, Legeai C. La greffe cardiaque en France: bilan actuel. *La Presse Médicale*. 2014;43:813–822. doi: 10.1016/j.lpm.2014.03.017
51. Agence De La Biomedecine. Greffe cardiaque. 2021. https://rams.agence-biomedecine.fr/sites/default/files/pdf/2022-08/ABM_PG_Organes_Coeur2021%2011.54.02.pdf
52. Dorent R, Jasseron C, Audry B, Bayer F, Legeai C, Cantrelle C, Smits JM, Eisen H, Jacquelinet C, Leprince P, et al. New French heart allocation system: comparison with Eurotransplant and US allocation systems. *Am J Transplant*. 2020;20:1236–1243. doi: 10.1111/ajt.15816
53. Cantrelle C, Jasseron C, Legeai C, Henaine R, Garaix F, Houyel L, Filipo SD, Kerbaul F, Dorent R. The New French Heart Allocation System has improved access to transplantation in adolescents. *J Heart Lung Transplant*. 2022;41:S173–S174. doi: 10.1016/j.healun.2022.01.410
54. Cowie MR. The heart failure epidemic: a UK perspective. *Echo Res Pract*. 2017;4:R15–R20. doi: 10.1530/ERP-16-0043
55. (NHSBT) NHSBaT. POL229/10 – heart transplantation: selection criteria and recipient registration. 2023. <https://nhsbtdeb.blob.core.windows.net/umbraco-assets-corp/29404/pol229.pdf>
56. Transplant NHSBa. Annual Activity report, section 7: cardiothoracic activity. 2023. <https://nhsbtdeb.blob.core.windows.net/umbraco-assets-corp/30141/section-7-cardiothoracic-activity.pdf>
57. Transplant NHSBa. Annual activity report. Section 11: Survival rates following transplantation. 2023. <https://nhsbtdeb.blob.core.windows.net/umbraco-assets-corp/30145/section-11-survival-rates-following-transplantation.pdf>
58. Transplant NHSBa. Annual report on cardiothoracic organ transplantation. Report for 2021/2022. 2022. <https://nhsbtdeb.blob.core.windows.net/umbraco-assets-corp/27816/nhsbt-annual-report-on-cardiothoracic-organ-transplantation-202122.pdf>
59. MacGowan GA, McDiarmid A, Jansen K, Coats L, Crossland D, Woods A, Kunadian V, Shah A, Schueler S, Parry G. Gender differences in the assessment, decision-making and outcomes for ventricular assist devices and heart transplantation: an analysis from a UK transplant center. *Clin Transplant*. 2022;36:e14666. doi: 10.1111/ctr.14666
60. Matskeplishvili S, Kontsevaya A. Cardiovascular health, disease, and care in Russia. *Circulation*. 2021;144:586–588. doi: 10.1161/CIRCULATIONAHA.121.055239
61. Kozlov IA, Krichevskiy LA. Heart transplantation in the USSR and former soviet countries. *J Cardiothorac Vasc Anesth*. 2020;34:3398–3408. doi: 10.1053/j.jvca.2019.12.001
62. Gautier SV, Khomyakov SM. Organ donation and transplantation in the Russian Federation in 2021. 14th Report from the Registry of the Russian Transplant Society. *Russian J Transplantol Artificial Organs*. 2022;24:8–31. doi: 10.15825/1995-1191-2022-3-8-31
63. Fomichev AVKDS, Agaeva HA, Zhulkov MO, Doronin DV, Chernyavsky AM. Experience of heart transplantation with an extended cold ischemic time of donor heart. *Russian J Cardiol*. 2020;25:4011. doi: 10.15829/1560-4071-2020-4011
64. Fedotov PA, Simonenko MA, Sazonova YV, Bortsova MA, Kostomarov AN, Sitnikova MY. Mortality risk factors in patients who are in heart transplantation waiting list. *OBM Transplant*. 2019;03:077. doi: 10.21926/1903077
65. Kalinina E. *Ensuring National Interest: Legal and Ethical Issues of Organ Transplantation in Russia*. European Proceedings of Social and Behavioural Sciences; 2021. doi: 10.15405/epsbs.2021.02.02.54
66. Osborn A. Hopes for Russia's transplant service rise as doctors are cleared of murder. *BMJ*. 2005;330:559. doi: 10.1136/bmj.330.7491.559-b
67. Eurotransplant. Annual Report. 2023. https://www.eurotransplant.org/wp-content/uploads/2024/06/ETP_AR2023_LowRes.pdf
68. Eurotransplant. Chapter 6: ET Thoracic Allocation System (ETHAS). 2022. <https://www.eurotransplant.org/wp-content/uploads/2024/11/H6-ETHAS-November-19-2024.pdf>
69. Planinc M, Sutlić Z, Rudež I, Barić D, Unić D, Blažeković R, Grubišić M, Varodić J, Starčević B, Planinc D. Results of heart transplantation: 18 years experience in University Hospital Dubrava, Zagreb. *J Cardiothorac Surg*. 2013;8:O151. doi: 10.1186/1749-8090-8-S1-O151
70. Mario Udovičić NBP, Mihovil S, Danijela G, Vanja Ivanović M, Hrvoje F, Igor R, Davor B, Daniel U, Robert B, Josip V, et al. Impact of recipient characteristics on survival after heart transplantation – a single-center experience. *Cardiol Croat*. 2022;17:202. doi: 10.15836/ccar2022.202
71. Katoue MG, Cerda AA, García LY, Jakovljevic M. Healthcare system development in the Middle East and North Africa region: challenges, endeavors and prospective opportunities. *Front Public Health*. 2022;10:1045739. doi: 10.3389/fpubh.2022.1045739
72. Berzon CI. 2008 organ transplant law: continued ethical challenges to the priority points model. *Isr J Health Policy Res*. 2018;7:11. doi: 10.1186/s13584-018-0203-6
73. Peled Y, Freimark D, Har-Zahav Y, Nachum E, Kogan A, Kassif Y, Lavee J. Trends and outcomes in heart transplantation over the past three decades: a single tertiary center experience in Israel. *Isr Med Assoc J*. 2018;20:567–572.
74. Hsieh EM. Matching the market for heart transplantation. *Circ Heart Fail*. 2016;9:e002679. doi: 10.1161/CIRCHEARTFAILURE.115.002679
75. Lavee J, Brock DW. Prioritizing registered donors in organ allocation: an ethical appraisal of the Israeli organ transplant law. *Curr Opin Crit Care*. 2012;18:707–711. doi: 10.1097/MCC.0b013e328357a2e2
76. Peled Y, Loewenthal R, Kassif Y, Raichlin E, Younis A, Younis A, Nachum E, Freimark D, Lavee J. Ethnic disparity in Israel impacts long-term results after heart transplantation. *Isr J Health Policy Res*. 2019;8:3. doi: 10.1186/s13584-018-0271-7

77. GBD 2017 Saudi Arabia Collaborators. The burden of disease in Saudi Arabia 1990–2017: results from the Global Burden of Disease Study 2017. *Lancet Planet Health*. 2020;4:e195–e208. doi: 10.1016/s2542-5196(20)30075-9
78. Khaliel F, Alshamadin F, Alsanei A, Alburaiqi J, Alhussein M, Saad E, Alhalees Z. Heart transplant challenges and long-term outcome in Saudi Arabia: single high-volume center. *J Heart Lung Transplant*. 2022;41:S241. doi: 10.1016/j.jhealun.2022.01.1745
79. AlHabeeb W, Al-Ayoubi F, AlGhalayini K, Al Ghofaili F, Al Hebaishi Y, Al-Jazairi A, Al-Mallah MH, AlMasood A, Al Qaseer M, Al-Saif S, et al. Saudi Heart Association (SHA) guidelines for the management of heart failure. *J Saudi Heart Assoc*. 2019;31:204–253. doi: 10.1016/j.jsha.2019.06.004
80. Selimovic N, Alhalees Z, Kjellman U, Alsanei A, Eyjolfsson A, Alhabeeb W, Alburaiqi J, Khouqeer F, Saad E, Khaliel F. Current status of heart transplantation in the kingdom of Saudi Arabia. *J Heart Lung Transplant*. 2014;33:S264–S265. doi: 10.1016/j.jhealun.2014.01.697
81. AlHabeeb W, AlAyoubi F, Tash A, AlAhmari L, AlHabib KF. Attitude of the Saudi community towards heart donation, transplantation, and artificial hearts. *Saudi Med J*. 2017;38:742–747. doi: 10.15537/smj.2017.7.18178
82. Ifeoma U, Chinwuba I, Ngozi I, Ejikeme A, Uchenna I, Julius O, Ugochi O, Chimezie O, Sani A, Obinna O. Organ donation and transplantation in Sub-Saharan Africa: opportunities and challenges. In: Vassil M, ed. *Organ Donation and Transplantation*. IntechOpen; 2020.
83. Loua A, Feroletto M, Sougou A, Kasilo OMJ, Nikiéma JB, Fuller WF, Kiazkov S, Tumusiime P. A review of policies and programmes for human organ and tissue donations and transplantations, WHO African Region. *Bull World Health Organ*. 2020;98:420–425. doi: 10.2471/BLT.19.236992
84. Shroff S, Mittal K, Navin S. Heart transplantation in India—looking back as we celebrate 25 years of the transplant law. *Indian J Thorac Cardiovasc Surg*. 2020;36:215–223. doi: 10.1007/s12055-020-00957-6
85. National Organ and tissue Transplant Organization. Allocation criteria for heart. 2016. <https://www.notto.gov.in/writereaddata/Portal/Images/pdf/29.02.2016Heart.pdf>
86. Balakrishnan K. Heart transplantation in India—are we there yet? *Indian J Thorac Cardiovasc Surg*. 2020;36:159–165. doi: 10.1007/s12055-020-00987-0
87. Vaidya G. Heart transplantation in India. *Indian Heart J*. 2021;73:518–520. doi: 10.1016/j.ihj.2021.06.013
88. Shi BY, Liu ZJ, Yu T. Development of the organ donation and transplantation system in China. *Chin Med J (Engl)*. 2020;133:760–765. doi: 10.1097/CM9.0000000000000779
89. National Health Commission. Notice on printing and distributing the basic principles and core policies of human organ distribution and sharing in China. 2018. <http://www.nhc.gov.cn/yzygj/s3586/201808/d35d96f2d-b82403ebe2ba41f2c583896.shtml>
90. Hu XJ, Dong NG, Liu JP, Li F, Sun YF, Wang Y. Status on heart transplantation in China. *Chin Med J (Engl)*. 2015;128:3238–3242. doi: 10.4103/0366-6999.170238
91. Wei X, Jiang H, Li Q. Organ transplantation in China—not yet a new era. *Lancet*. 2014;384:741. doi: 10.1016/s0140-6736(14)61429-x
92. Sharif A. Why journals should stop publishing transplantation research from China. *BMJ*. 2021;373:n1069. doi: 10.1136/bmj.n1069
93. Robertson MP, Hinde RL, Lavee J. Analysis of official deceased organ donation data casts doubt on the credibility of China's organ transplant reform. *BMC Med Ethics*. 2019;20:79. doi: 10.1186/s12910-019-0406-6
94. The Lancet Regional Health-Western Pacific. South Korea's population shift: challenges and opportunities. *Lancet Reg Health West Pac*. 2023;36:100865. doi: 10.1016/j.lanwpc.2023.100865
95. Park J, Kim CJ. Recent decrease in organ donation from brain-dead potential organ donors in Korea and possible causes. *J Korean Med Sci*. 2020;35:e94. doi: 10.3346/jkms.2020.35.e94
96. Kim IC, Youn JC, Kobashigawa JA. The past, present and future of heart transplantation. *Korean Circ J*. 2018;48:565–590. doi: 10.4070/kcj.2018.0189
97. Kim D, Choi JO, Oh J, Cho HJ, Jung SH, Lee HY, Kang SM, Kim JJ, Jeon ES. The Korean Organ Transplant Registry (KOTRY): second official adult heart transplant report. *Korean Circ J*. 2019;49:724–737. doi: 10.4070/kcj.2018.0392
98. Colvin MM, Smith JM, Ahn YS, Messick E, Lindblad K, Israni AK, Snyder JJ, Kasiske BL. OPTN/SRTR 2021 annual data report: heart. *Am J Transplant*. 2023;23:S300–S378. doi: 10.1016/j.ajt.2023.02.008
99. Zealand TTSOA. Clinical guidelines for organ transplantation from deceased donors. 2022. https://tsanz.com.au/storage/documents/TSANZ_Clinical_Guidelines_Version-19_15062022.pdf
100. The Transplantation Society of Australia and New Zealand. Clinical guidelines for organ transplantation from deceased donors. version 1.10. https://tsanz.com.au/storage/documents/TSANZ_Clinical_Guidelines_Version-110_Final.pdf
101. ANZOD. 2022 Annual Report, section 7: deceased donor heart donation. AUSTRALIA and New Zealand dialysis and transplant registry, Adelaide, Australia. 2022. https://www.anzdata.org.au/wp-content/uploads/2022/05/s07_heart_2021_ar_2022_section_v1.0_20220520_final.pdf
102. Dhital K, Ludhani P, Scheuer S, Connellan M, Macdonald P. DCD donations and outcomes of heart transplantation: the Australian experience. *Indian J Thorac Cardiovasc Surg*. 2020;36:224–232. doi: 10.1007/s12055-020-00998-x
103. Chew HC, Iyer A, Connellan M, Scheuer S, Villanueva J, Gao L, Hicks M, Harkness M, Soto C, Dinale A, et al. Outcomes of donation after circulatory death heart transplantation in Australia. *J Am Coll Cardiol*. 2019;73:1447–1459. doi: 10.1016/j.jacc.2018.12.067
104. Czekajewska JM, Jaworowicz-Zimny A. Organ transplant in present-day Japan: reasons behind low numbers of deceased donors. *Diametros*. 2021;18:1–24. doi: 10.33392/diam.1630
105. Kitamura S. Heart transplantation in Japan: a critical appraisal for the results and future prospects. *Gen Thorac Cardiovasc Surg*. 2012;60:639–644. doi: 10.1007/s11748-012-0110-7
106. Japan Organ Transplant Network. Heart transplant candidate (recipient) selection criteria. 2015. <https://www.jotnw.or.jp/files/page/medical/manual/doc/rec-heart.pdf>
107. The Japanese Society for Heart Transplantation. The registry report of Japanese Heart Transplantation—2021. *Jpn J Transplant*. 2021;56:235–244. doi: 10.11386/jst.56.3_235
108. Evans RW. Socioeconomic aspects of heart transplantation. *Curr Opin Cardiol*. 1995;10:169–179. doi: 10.1097/00001573-199503000-00012
109. The declaration of Istanbul on organ trafficking and transplant tourism. *Indian J Nephrol*. 2008;18:135–140. doi: 10.4103/0971-4065.43686
110. Assounga A. The Abuja declaration on cells, tissue and organ donation and transplantation. *Afr J Nephrol*. 2012;15:6–8. doi: 10.21804/15-1-748
111. Martin DE, Van Assche K, Domínguez-Gil B, López-Fraga M, García Gallont R, Muller E, Capron AM. Strengthening global efforts to combat organ trafficking and transplant tourism: implications of the 2018 edition of the declaration of Istanbul. *Transplant Direct*. 2019;5:e433. doi: 10.1097/TXD.0000000000000872